September 11, 2026

The Editors
Computers & Industrial Engineering

Dear Editors,

Please consider our manuscript, "Completion-aware search with partial witness repair for dense fixed-shape facility layout," for publication as a research article in *Computers & Industrial Engineering*.

Dense layout design calls for search decisions that preserve room for the remaining facilities. Our paper uses a known feasible layout as reusable geometry for subsequent construction. We ask when this information improves solution quality under the same time budget, connecting constructive search with the use of existing solutions in neighborhood design.

We develop a partial witness repair rule. After a proposed placement, the method repacks only the displaced facilities and retains all unaffected rectangles in the stored completion. Building on the feasible-continuation principle of fortified rollout, this rule provides a targeted repair neighborhood for tightly packed layouts. We keep the compatible completion separate from the best full layout, and independently verify every candidate before it can update the returned solution.

The main study compares partial repair, full completion regeneration, and a validation-selected adaptive repair control on 240 attempted instances with 32 to 256 facilities, two geometry families, and fills of 90% and 95%. The common initializer succeeds on 236 instances, yielding 2,124 runs. At 95% fill, partial repair exceeds the adaptive control by 3.45 to 5.91 percentage points in initial-objective improvement at 10 seconds and 1.68 to 4.12 points at 60 seconds. With 256 facilities, partial repair gives higher quality than full regeneration at 10 seconds, while full regeneration leads at 60 seconds. A separate one-second study isolates the benefit of nonempty repair beyond unchanged witness reuse. Together, these results show when targeted repair offers an early improvement and when broader reconstruction benefits from additional time.

The paper addresses a computational decision problem in industrial engineering: improving a feasible facility arrangement while preserving hard packing constraints. Its contribution is a concrete repair mechanism and a controlled evaluation of when retaining a feasible completion is worth its computational cost. We believe this will interest readers of *Computers & Industrial Engineering* working on facility layout, computer-aided planning and heuristic optimization.

The source code, benchmark generators, result files, and independent verification scripts are publicly available at `https://github.com/yeoneung/layout_optimization`. The Online Supplement provides the complete budget comparisons and additional diagnostics.

Thank you for considering our manuscript.

Sincerely,
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